

ECO 310 Problem Set 2 - Problem 4 Solution

Risk Aversion and Speeding Tickets

Problem Statement

You have probability p of getting a speeding ticket that costs ϕ . The government wishes to hold expected speeding ticket revenue $p\phi = R \geq 0$ constant.

Question: If you are strictly risk-averse, does a reduction in p (subject to $p\phi = R$) help you or hurt you?

Given hint: For a risk-averse consumer with concave u , for any $a < b$:

$$u(b) - u(a) > u'(a)(b - a)$$

Solution

Step 1: Write out the Expected Utility

You start with initial wealth w_0 and face two possible outcomes:

- With probability $(1 - p)$: no ticket, final wealth = w_0
- With probability p : ticket, final wealth = $w_0 - \phi$

Expected utility is:

$$EU = (1 - p)u(w_0) + p \cdot u(w_0 - \phi)$$

Since the constraint is $p\phi = R$, we can substitute $\phi = \frac{R}{p}$:

$$EU(p) = (1 - p)u(w_0) + p \cdot u\left(w_0 - \frac{R}{p}\right)$$

Step 2: Differentiate EU with respect to p

To see whether reducing p helps or hurts, we need to find $\frac{dEU}{dp}$:

$$\begin{aligned}\frac{dEU}{dp} &= \frac{d}{dp} \left[(1 - p)u(w_0) + p \cdot u\left(w_0 - \frac{R}{p}\right) \right] \\ &= -u(w_0) + u\left(w_0 - \frac{R}{p}\right) + p \cdot u'\left(w_0 - \frac{R}{p}\right) \cdot \frac{R}{p^2} \\ &= -u(w_0) + u\left(w_0 - \frac{R}{p}\right) + u'\left(w_0 - \frac{R}{p}\right) \cdot \frac{R}{p}\end{aligned}$$

Step 3: Simplify using substitution

Let:

$$a = w_0 - \frac{R}{p} \quad (\text{wealth if you get a ticket})$$

$$b = w_0 \quad (\text{wealth if you don't get a ticket})$$

Note that $a < b$ since $\frac{R}{p} > 0$. Also, $b - a = \frac{R}{p}$.

Substituting:

$$\begin{aligned} \frac{dEU}{dp} &= -u(b) + u(a) + u'(a) \cdot (b - a) \\ &= u(a) - u(b) + u'(a)(b - a) \\ &= -[u(b) - u(a) - u'(a)(b - a)] \end{aligned}$$

Step 4: Apply risk aversion (concavity)

For a risk-averse consumer with concave u , we know that for $a < b$:

$$u(b) - u(a) > u'(a)(b - a)$$

This says the actual utility increase is greater than the linear approximation (the function curves below its tangent line due to decreasing marginal utility).

Therefore:

$$u(b) - u(a) - u'(a)(b - a) > 0$$

Step 5: Conclude

Since $u(b) - u(a) - u'(a)(b - a) > 0$, we have:

$$\frac{dEU}{dp} = -[u(b) - u(a) - u'(a)(b - a)] < 0$$

Because $\frac{dEU}{dp} < 0$, expected utility **decreases** as p increases. This means expected utility **increases** as p decreases.

Answer

A reduction in p HELPS you (increases your expected utility) if you are strictly risk-averse.

Intuition

Even though the expected cost $p\phi = R$ remains constant, the distribution of risk matters:

- **Higher p , lower ϕ :** More likely to pay, but smaller fine
- **Lower p , higher ϕ :** Less likely to pay, but larger fine

A risk-averse person prefers spreading risk over many small losses rather than concentrating it in fewer large losses. Therefore, increasing p (more frequent, smaller fines) is actually better than decreasing p (less frequent, larger fines), even though the expected payment is the same.